The documentation and process conversion measures necessary to comply with this document shall be completed by 3 September 2008.

INCH-POUND

MIL-PRF-19500/633C
3 June 2008
SUPERSEDING
MIL-PRF-19500/633B
16 March 2004

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTORS, P-CHANNEL
SILICON, TYPES 2N7403 AND 2N7404, JANSD AND JANSR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a P-channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event characterization), power transistor. One level of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-254).

1.3 Maximum ratings. $T_A$ = +25°C, unless otherwise specified.

| Type | $P_T$ (1) $T_C$ = +25°C | $P_T$ $T_A$ = +25°C | $V_{DS}$ | $V_{DG}$ | $V_{GS}$ | $R_{\theta JC}$ Max | $I_{D1}$ (2) (3) $T_C$ =+25°C | $I_{D2}$ $T_C$ = +100°C | $I_S$ (2) | $I_{DM}$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | V dc | V dc | V dc | °C/W | A dc | A dc | A dc | A (pk) | °C |
| 2N7403 | 125 | 50 | -100 | -100 | ±20 | 1.00 | 22 | 14 | 22 | 66 | -55 to +150 |
| 2N7404 | | | -200 | -200 | ±20 | 1.00 | 15 | 9 | 15 | 45 | |

(1) Derate linearly 1.0 W/°C for $T_C$ > +25°C;
(2) The following formula derives the maximum theoretical $I_D$ limit. $I_D$ is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

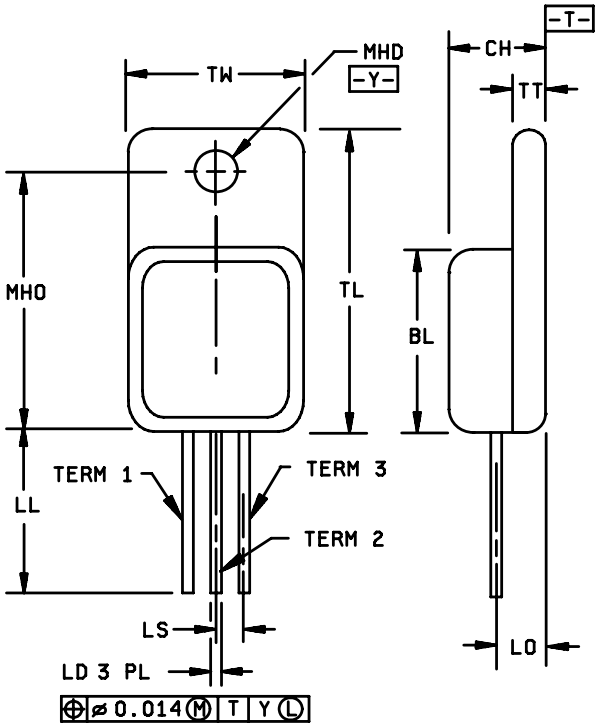
(3) See figure 2, maximum drain current graphs.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

AMSC N/A

FSC 5961

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| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .530 | .550 | 13.46 | 13.97 |
| CH | .249 | .260 | 6.32 | 6.60 |
| LD | .035 | .045 | 0.89 | 1.14 |
| LL | .510 | .570 | 12.95 | 14.48 |
| LO | .150 BSC | | 3.81 BSC | |
| LS | .150 TYP | | 3.81 TYP | |
| MHD | .139 | .149 | 3.53 | 3.78 |
| MHO | .665 | .685 | 16.89 | 17.40 |
| TL | .790 | .800 | 20.07 | 20.32 |
| TT | .040 | .050 | 1.02 | 1.27 |
| TW | .535 | .545 | 13.59 | 13.84 |
| Term 1 | Drain | | | |
| Term 2 | Source | | | |
| Term 3 | Gate | | | |

NOTES:
1. Dimensions are in millimeters.
2. Millimeters are given for general information only.
3. All terminals are isolated from case.
4. Die to base is BeO isolated, terminals to case ceramic ($AL_2O_3$) isolated.
5. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

* FIGURE 1. Physical dimensions (similar to TO-254).



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1.4 Primary electrical characteristics at $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 $I_D$ = 1.0 mA dc | $V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ $I_D$ = 1.0 mA dc | Max $I_{DSS1}$ $V_{GS}$ = 0 $V_{GS}$ = 80 percent of rated $V_{DS}$ | Max $r_{DS(on)}$ (1) $V_{GS}$ = -12 V | | $I_{AS} = I_{DM}$ |
|---|---|---|---|---|---|---|
| | | | | $T_J$ = 25°C at $I_{D2}$ | $T_J$ = 125°C at $I_{D2}$ | |
| | V dc | V dc Min Max | µA dc | Ω | Ω | A (pk) |
| 2N7403 | -100 | -2.0 -6.0 | 25 | 0.140 | 0.217 | 66 |
| 2N7404 | -200 | | | 0.290 | 0.513 | 45 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

nC .....nano coulomb.
$I_{AS}$ .....Rated avalanche current, non-repetitive.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (similar to TO-254) herein.

3.4.1 Lead finish and material. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.7.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq$ or 100 kΩ, whenever bias voltage is applied drain to source.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor but shall be retained on the initial container.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.



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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with MIL-PRF-19500.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein shall be performed on the first inspection lot of this revision to maintain qualification.



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4.3 Screening (JANS only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) (1) (2) | Measurement |
|---|---|
| | JANS |
| (3) | Gate stress test (see 4.3.1) |
| (3) | Method 3470 of MIL-STD-750, (see 4.3.2) |
| (3) 3c | Method 3161 of MIL-STD-750, (see 4.3.3) |
| 7 | Optional. |
| 9 | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, subgroup 2 of table I herein. |
| 10 | Method 1042 of MIL-STD-750, test condition B |
| 11 | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(ON)1}$, $V_{GS(TH)1}$<br>Subgroup 2 of table I herein.<br><br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 μA dc or ±100 percent of initial value, whichever is greater. |
| 12 | Method 1042 of MIL-STD-750, test condition A |
| 13 | Subgroups 2 and 3 of table I herein<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 μA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(ON)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(TH)1}$ = ±20 percent of initial value. |
| 14 | Required. |

(1) At the end of the test program, $I_{GSSF1}$, $I_{GSSR1}$ and $I_{DSS1}$, are measured.
(2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime before screen 9.



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4.3.1 Gate stress test. Apply $V_{GS}$ = -30 V minimum for t = 250 µs minimum.

4.3.2 Single pulse avalanche energy ($E_{AS}$).

a. $I_{AS} = I_{DM.}$

b. L = 0.1 mH.

c. $E_{AS} = 1/2\ LI_{AS}^2$.

d. $V_{DD}$ = -50 V to -150 V dc.

e. Initial junction temperature = 25°C, -5°C, +10°C.

4.3.3 Thermal impedance ($\Delta V_{SD}$ measurement). The delta $V_{SD}$ measurement shall be performed in accordance with method 3161 of MIL-STD-750. The delta $V_{SD}$ conditions ($I_H$ and $V_H$) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 3) and shall be specified in the certificate of conformance prior to qualification. The following parameter measurements shall apply:

| | |
|---|---|
| a. Measuring current ($I_M$) | 10 mA |
| b. Drain heating current ($I_H$) | 4 A |
| c. Heating time ($t_H$) | 100 ms |
| d. Drain-source heating voltage ($V_H$) | -25 V |
| e. Measurement time delay ($t_{MD}$) | 30 - 60 µs |
| f. Sample window time ($t_{SW}$) | 10 µs maximum |

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIa (JANS) of MIL-PRF-19500, and as follows. End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1051 | Test condition G, 100 cycles |
| B3 | 2077 | SEM |
| B4 | 1042 | Intermittent operation life, condition D, 2,000 cycles. No heat sink nor or forced-air Cooling on the device shall be permitted during the on cycle. $t_{ON}$ = 30 seconds minimum. |
| B5 | 1042 | Accelerated steady-state reverse bias, condition A, $V_{DS}$ = rated; $T_A$ = +175°C; t = 120 hours, minimum. |
| B5 | 1042 | Accelerated steady-state gate bias, condition B, $V_{GS}$ = rated; $T_A$ = +175°C; t = 24 hours. |
| B5 | 2037 | Bond strength, test condition A |
| B6 | 3161 | Thermal resistance, see 4.5.2. |



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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Terminal strength, test condition A, weight = 10 lbs., t = 15 sec. |
| C5 | 3161 | See 4.5.2. |
| C6 | 1042 | Test condition D, 6,000 cycles; 1 cycle = 30 sec. min. |

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein.

4.4.4.1 Design parameters. Not tested on a per lot basis. Design shall be such that the devices shall be capable of meeting the requirements herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC}$ = 1.0°C/W. The following parameters shall apply:

a. Measuring current($I_M$) ..... 10 mA.

b. Drain heating current ($I_H$) ..... 4 A.

c. Heating time ($t_H$) ..... Steady-state (see method 3161 of MIL-STD-750).

d. Drain-source heating voltage ($V_H$) ..... -25 V.

e. Measurement time delay ($t_{MD}$) ..... 30 to 60 μs.

f. Sample window time ($t_{SW}$)..... 10 μs maximum.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| * Thermal impedance | 3161 | See 4.3.2 | | | | |
| Breakdown voltage, drain to source | 3407 | Bias condition C<br>$V_{GS}$ = 0 V, $I_D$ = 1.0 mA dc | $V_{(BR)DSS}$ | | | |
| 2N7403 | | | | -100 | | V dc |
| 2N7404 | | | | -200 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DG} \geq V_{GS}$, | $V_{GS(th)1}$ | -2.0 | -6.0 | V dc |
| Gate current | 3411 | Bias condition C, $V_{GS}$ = +20 V dc, $V_{DS}$ = 0 | $I_{GSSF1}$ | | +100 | nA dc |
| Gate current | 3411 | Bias condition C, $V_{GS}$ = -20 V dc, $V_{DS}$ = 0 | $I_{GSSR1}$ | | -100 | nA dc |
| Drain current | 3413 | Bias condition C, $V_{GS}$ = 0 V dc, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS1}$ | | 25 | µA dc |
| Static drain to source on-state resistance | 3421 | Bias condition A, $V_{GS}$ = -12 V dc, $I_D$ = $I_{D2}$, pulsed (see 4.5.1) | $r_{DS(ON)1}$ | | | |
| 2N7403 | | | | | 0.14 | Ω |
| 2N7404 | | | | | 0.29 | Ω |
| Static drain to source on-state voltage | 3405 | Bias condition A, $V_{GS}$ = -12 V dc, $I_D$ = $I_{D2}$, pulsed (see 4.5.1) | $V_{DS(ON)1}$ | | | |
| 2N7403 | | | | | -3.23 | V dc |
| 2N7404 | | | | | -4.57 | V dc |
| Forward voltage | 4011 | Pulsed (see 4.5.1), $I_D$ = $I_{D1}$<br>$V_{GS}$ = 0 V dc | $V_{SD}$ | -0.6 | -1.8 | V dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C$ = $T_J$ = +125°C | | | | |
| Gate current | 3411 | Bias condition C, $V_{GS}$ = ±20 V dc, $V_{DS}$ = 0 V dc, | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | Bias condition C, $V_{GS}$ = 0 V dc, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS2}$ | | 0.25 | mA dc |

See footnote at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 3 - Continued | | | | | | |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = -12 V dc, pulsed (see 4.5.1), $I_D$ = rated $I_{D2}$ | $r_{DS(on)2}$ | | | |
| 2N7403 | | | | | 0.217 | Ω |
| 2N7404 | | | | | 0.513 | Ω |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA | $V_{GS(th)2}$ | -1.0 | | V dc |
| Low temperature operation: | | $T_C = T_J$ = -55°C | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA | $V_{GS(th)3}$ | | -7.0 | V dc |
| Subgroup 4 | | | | | | |
| Switching time test | 3472 | $I_D$ = rated $I_{D1}$; $V_{GS}$ = -12 V dc; $R_G$ = 4.7 Ω; $V_{DD}$ = 50 percent of rated $V_{DS}$ | | | | |
| Turn-on delay time | | | $t_{d(on)}$ | | | |
| 2N7403 | | | | | 110 | ns |
| 2N7404 | | | | | 120 | ns |
| Rise time | | | $t_r$ | | | |
| 2N7403 | | | | | 390 | ns |
| 2N7404 | | | | | 160 | ns |
| Turn-off delay time | | | $t_{d(off)}$ | | | |
| 2N7403 | | | | | 300 | ns |
| 2N7404 | | | | | 280 | ns |
| Fall time | | | $t_f$ | | | |
| 2N7403 | | | | | 170 | ns |
| 2N7404 | | | | | 120 | ns |

See footnote at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area test (high voltage) | 3474 | See figure 4, $t_p$ = 10 ms, $V_{DS}$ = 80 percent of rated $V_{DS}$, $V_{DS} \leq$200 V dc max. | | | | |
| Electrical measurements | | See table IV, steps 1, 2, 3, 4, 5, 6, and 7 | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | Condition B | | | | |
| On-state gate charge | | | $Q_{g(on)}$ | | | |
| 2N7403 | | | | | 160 | nC |
| 2N7404 | | | | | 150 | nC |
| Gate to source charge | | | $Q_{gs}$ | | | |
| 2N7403 | | | | | 29 | nC |
| 2N7404 | | | | | 32 | nC |
| Gate to drain charge | | | $Q_{gd}$ | | | |
| 2N7403 | | | | | 65 | nC |
| 2N7404 | | | | | 67 | nC |
| Reverse recovery time | 3473 | $d_i/d_t \leq$ 100 A/µs, $V_{DD} \leq$ 30 V, | $t_{rr}$ | | | |
| 2N7403 | | | | | 270 | ns |
| 2N7404 | | | | | 300 | ns |

1/ For sampling plan, see MIL-PRF-19500.



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TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ 4/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | Post irradiation limits | | Units |
|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | Min. | Max. | Min. | Max. | |
| Subgroup 1 | | | | | | | | |
| Not applicable | | | | | | | | |
| Subgroup 2 | | $T_C$ = +25°C | | | | | | |
| Steady state total dose irradiation ($V_{GS}$ bias) | 1019 | $V_{GS}$ = -12 V, $V_{DS}$ = 0 V | | | | | | |
| Steady state total dose irradiation ($V_{DS}$ bias) | 1019 | $V_{GS}$ = 0 V, $V_{DS}$ = 80 percent of rated $V_{DS}$ | | | | | | |
| Breakdown voltage drain to source | 3407 | $V_{GS}$ = 0 V, $I_D$ = 1 mA dc, bias condition C | $V_{(BR)DSS}$ | | | | | |
| 2N7403 | | | | -100 | | -100 | | V dc |
| 2N7404 | | | | -200 | | -200 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)1}$ | -2.0 | -6.0 | -2.0 | -6.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = ±20 V dc, $V_{DS}$ = 0 V, bias condition C | $I_{GSS1}$ | | ±100 | | ±100 | |
| Drain current | 3413 | $V_{GS}$ = 0V, $V_{DS}$ = 80 percent of rated $V_{DS}$, bias condition C | $I_{DSS1}$ | | 25 | | 25 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ =12V dc, condition A, pulsed (see4.5.1), $I_D$ = $I_{D2}$ | $r_{DS(ON)1}$ | | | | | |
| 2N7403 | | | | | 0.14 | | 0.14 | Ω |
| 2N7404 | | | | | 0.29 | | 0.29 | Ω |
| Static drain to source on-state voltage | 3405 | $V_{GS}$ = -12V dc, condition A, pulsed (see 4.5.1), $I_D$ = $I_{D1}$ | $V_{DS(ON)}$ | | | | | |
| 2N7403 | | | | | -3.23 | | -3.23 | V dc |
| 2N7404 | | | | | -4.57 | | -4.57 | V dc |

1/ For sampling plan see MIL-PRF-19500.
2/ Electrical specifications are for 'D' and 'R' rad levels.
3/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification utilizing the same die design.
4/ At the manufacturers option, group D samples need not be subjected to all the screening tests, but shall be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.



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* TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

| Inspection 1/ 2/ 3/ 4/ 5/ | MIL-STD-750 | | Qualification and large lot conformance inspection. |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 12 devices<br>c = 0 |
| Temperature cycle | 1051 | Condition G, 500 cycles | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8 | |
| Subgroup 2 1/ | | | 45 devices<br>c = 0 |
| Steady-state reverse bias | 1042 | Condition A, 1,000 hours | |
| Electrical measurements | | See table IV, steps 1, 2, 3, 4, 5, 6, and 7 | |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours | |
| Electrical measurements | | See table IV, steps 1, 2, 3, 4, 5, 6, and 7 | |
| Subgroup 3 | | | 3 devices,<br>c = 0 |
| DPA | 2102 | | |
| Subgroup 4 | | | Sample size<br>N/A |
| Thermal impedance curves | | Each supplier shall submit their (typical) design maximum thermal impedance curves. In addition, the optimal test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 3 devices |
| ESD | 1020 | | |
| Subgroup 8 | | | 22 devices<br>c = 0 |
| Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | | |

See footnotes at end of table.



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* TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only - Continued.

| Inspection 1/ 2/ 3/ 4/ 5/ | MIL-STD-750 | | Qualification and large lot conformance inspection. |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 9 | | | 3 devices 5/<br>c = 0 |
| Electrical measurements 3/ | | See table IV, steps 3 and 4 | |
| See effect testing 4/ | 1080 | See figure 5<br>Fluence = 3e5 ±20 percent ions/$cm^2$<br>Flux = 5e3 to 2e4 ions/$cm^2$sec<br>Beam energy = 260 to 300 MeV<br>Temperature = 25°C ± 5°C | |
| 2N7403 | | LET = 26 to 30 MeV-$cm^2$/mg<br>Ion range = 40 to 45 microns<br>Insitu bias conditions: $V_{DS}$ = -100 V and $V_{GS}$ = 20 V<br><br>LET = 36 to 40 MeV-$cm^2$/mg<br>Ion range = 35 to 40 microns<br>Insitu bias conditions: $V_{DS}$ = -100 V and $V_{GS}$ = 10 V<br>$V_{DS}$ = -80 V and $V_{GS}$ = 15 V<br>$V_{DS}$ = -50 V and $V_{GS}$ = 20 V | |
| 2N7404 | | LET = 26 to 30 MeV-$cm^2$/mg<br>Ion range = 40 to 45 microns<br>Insitu bias conditions: $V_{DS}$ = -200 V and $V_{GS}$ = 20 V<br><br>LET = 36 to 40 MeV-$cm^2$/mg<br>Ion range = 35 to 40 microns<br>Insitu bias conditions: $V_{DS}$ = -200 V and $V_{GS}$ = 5 V<br>$V_{DS}$ = -160 V and $V_{GS}$ = 10 V<br>$V_{DS}$ = -100 V and $V_{GS}$ = 15 V<br>$V_{DS}$ = -40 V and $V_{GS}$ = 20 V | |
| Electrical measurements 3/ | | See table IV, steps 3 and 4 | |

1/ A separate sample for each test may be pulled.
2/ Group E qualification of single event effect testing may be performed prior to lot formation. Wafers qualified to these group E QCI requirements may be used for any other specification utilizing the same die design.
3/ As a minimum, gate to source leakage and drain to source leakage are to be examined to verify the electrical performance of the DUT prior to and after test. At the manufacturer's option, the remaining static tests in table IV, with the exception of step 8, may be performed.
4/ Devices passing a given combination of drain and gate voltage for an LET of 36 to 40 MeV-$cm^2$/mg qualify the same conditions for an LET of 26 to 30 MeV-$cm^2$/mg.
5/ This sampling plan applies to each bias condition defined.



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TABLE IV. Group A, B, C and E electrical and delta measurements

| Step | Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limits | | Units |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Breakdown voltage drain to source<br>2N7403<br>2N7404 | 3407 | Bias condition C, $V_{GS}$ = 0 V, $I_D$ = 1 mA dc | $V_{(BR)DSS}$ | <br>-100<br>-200 | | <br>V dc<br>V dc |
| 2. | Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)1}$ | -2.0 | -6.0 | V dc |
| 3. | Gate current | 3411 | $V_{GS}$ = +20 V and -20 V dc, Bias condition C, $V_{DS}$ = 0 V | $I_{GSS1}$ | | ±100 | nA dc |
| 4. | Drain current | 3413 | Bias condition C, $V_{GS}$ = 0 V dc, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS1}$ | | 25 | µA dc |
| 5. | Static drain to source on-state resistance<br>2N7403<br>2N7404 | 3421 | $V_{GS}$ = -12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D2}$ | $r_{DS(ON)1}$ | | <br>0.14<br>0.29 | <br>Ω<br>Ω |
| 6. | Static drain to source on-state voltage<br>2N7403<br>2N7404 | 3405 | Pulsed (see 4.5.1), $V_{GS}$ = -12 V dc, condition A, $I_D = I_{D1}$ | $V_{DS(ON)}$ | | <br>-3.23<br>-4.57 | <br>V dc<br>V dc |
| 7. | Forward voltage | 4011 | $V_{GS}$ = 0 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D1}$ | $V_{SD}$ | -0.6 | -1.8 | V dc |
| 8. | Thermal impedance | 3161 | See 4.3.3 | $\Delta V_{SD}$ | | 136 | mV |

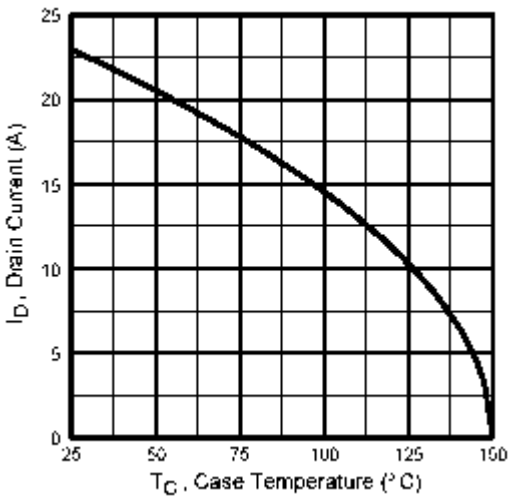
1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
   b. Subgroup 4, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.
   c. Subgroup 5, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.

2/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2 and 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
   b. Subgroup 6, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

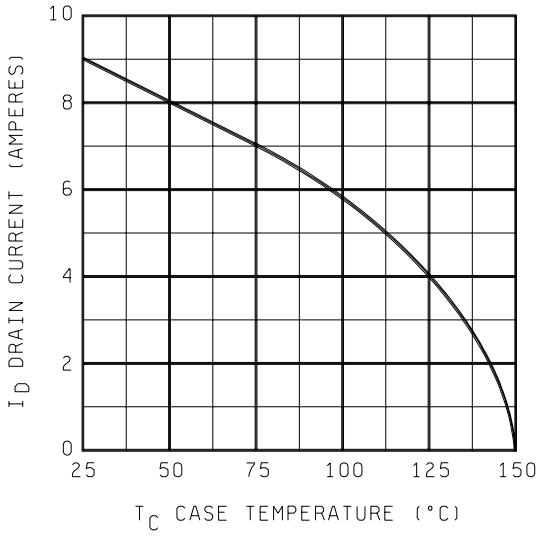
3/ The electrical measurements for table IX of MIL-PRF-19500 are as follows:
   a. Subgroup 1, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.
   b. Subgroup 2, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
   c. Subgroup 8, see table IV herein, steps 3 and 4.



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2N7403




2N7404

FIGURE 2. Maximum drain current vs case temperature graphs.



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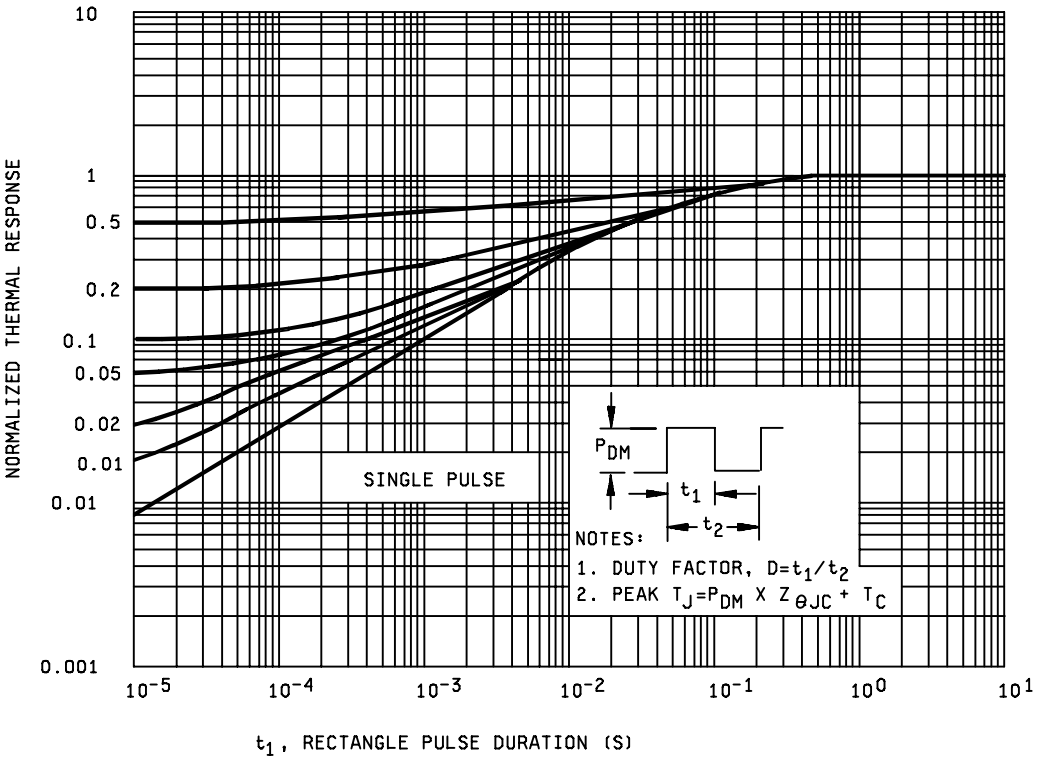



FIGURE 3. Thermal response curves.



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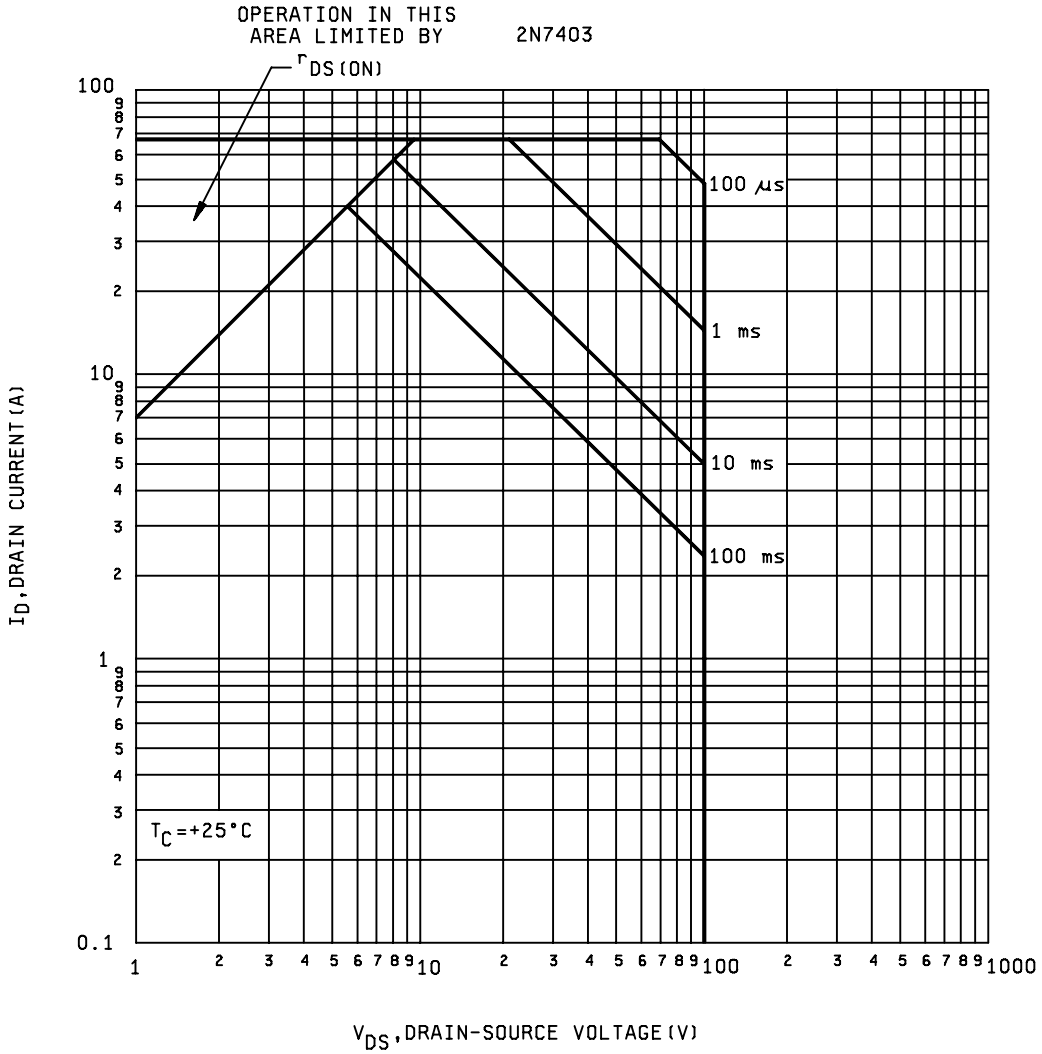



FIGURE 4. Safe operating area graphs.



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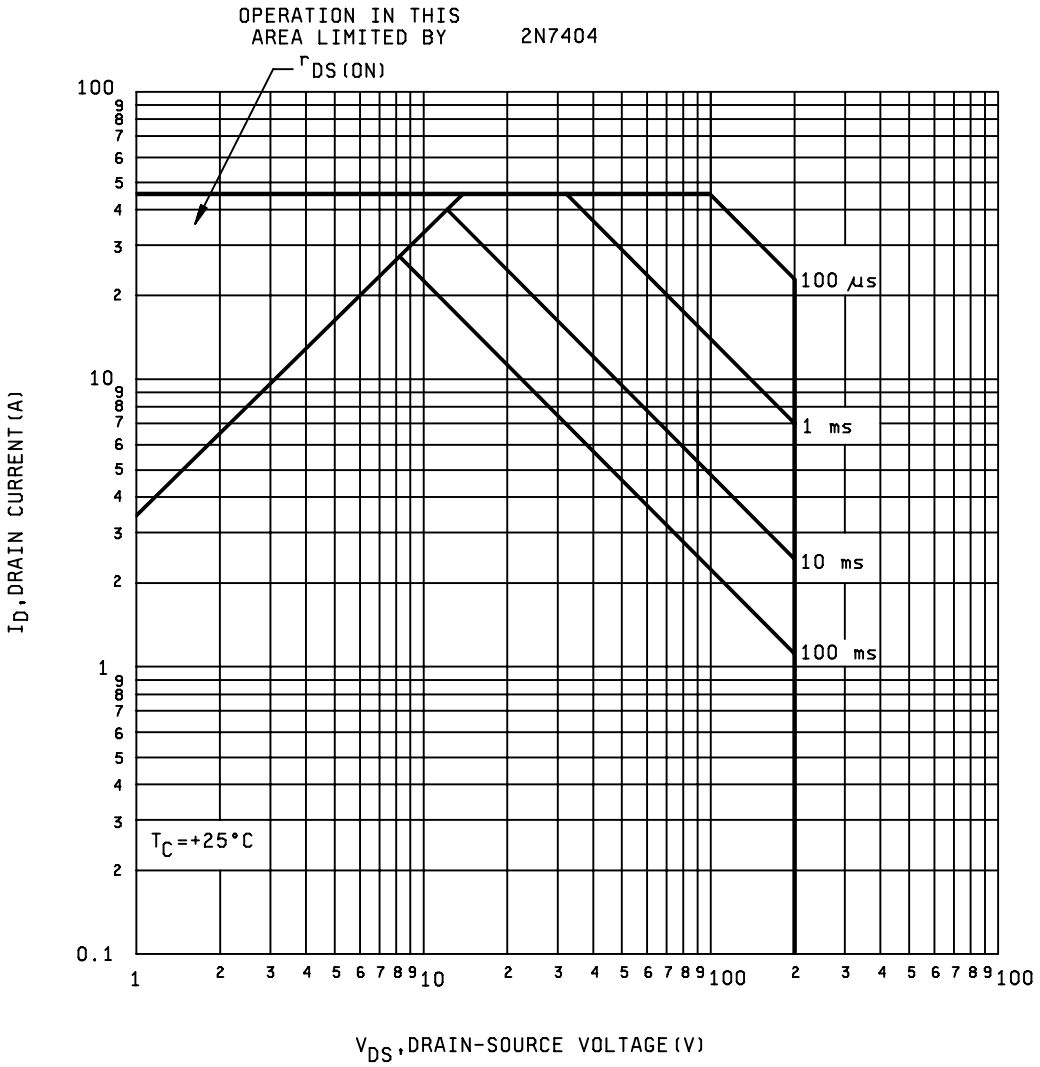



FIGURE 4. Safe operating area graphs - Continued.



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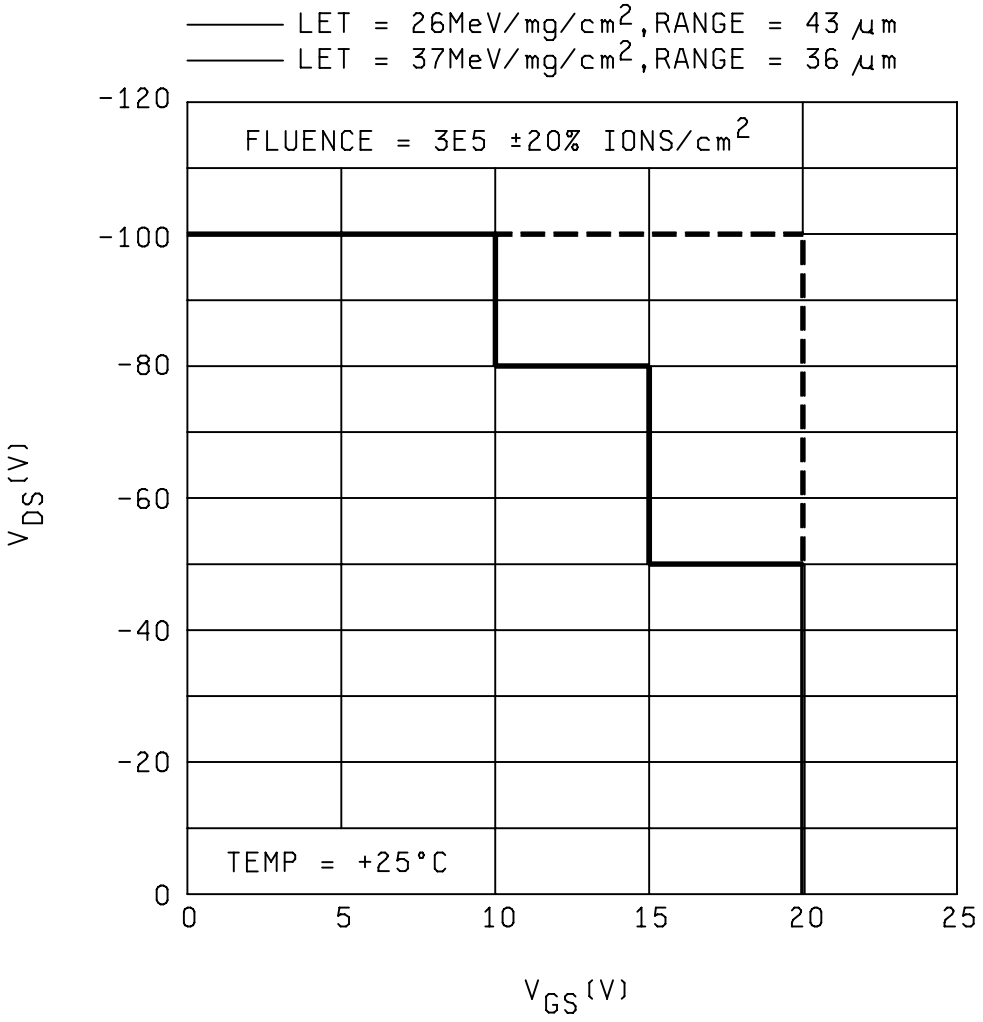



FIGURE 5. Single event effects safe operating area graphs.



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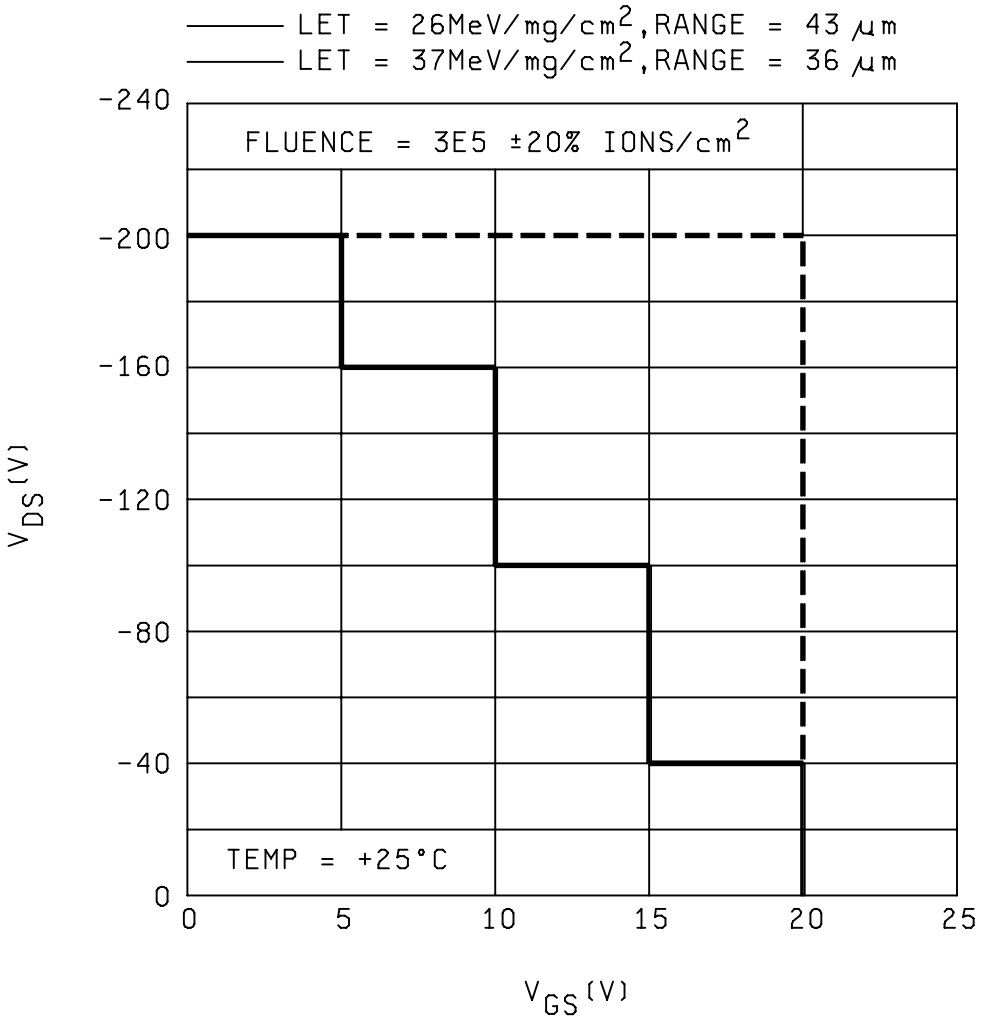



FIGURE 5. Single event effects safe operating area graphs - Continued



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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Lead material and finish (see 3.4.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML No. 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://assist.daps.dla.mil .

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PINs are subitable for the military PIN.

| Generic PIN | Military PIN |
|---|---|
| SF9150 | 2N7403 |
| SF9250 | 2N7404 |



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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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